

K1432 — FALSIFIER SCRUB (#107), sharpest live falsifier: the neutrino mass sum. ★ The corpus number is STALE: BST_NeutrinolessDoubleBeta.md cites “the DESI+Planck bound (0.072 eV)”; current (2025–26) is DESI DR2 + CMB < 0.064 eV (95%, Λ CDM), with the tightest combos (DESI + CMB + late-time, no H_0 prior) at < 0.050 eV (2σ) and the relaxed/dynamical-DE combos at < 0.10 – 0.12 eV. ★ BST predicts $\Sigma m_\nu = 0.058$ eV ($m_1=0$ exactly, $m_2=0.00865$, $m_3=0.04940$, normal ordering). Status: SAFE by a hair under standard DESI-DR2 Λ CDM ($0.058 < 0.064$); IN TENSION under the most aggressive Λ CDM combo ($0.058 > 0.050$); COMFORTABLE under dynamical DE (< 0.10 – 0.12). ★ TWO honest flags: (1) BST’s $\Sigma = 0.058$ IS the generic minimal-NO floor (0.0582) to 0.3% — NOT a distinctive BST prediction; any $m_1=0$ normal-ordering theory gives this. Don’t sell the *sum*; the distinctive content is $m_1=0$ *exactly* + the specific m_2/m_3 + the $0\nu\beta\beta$ window [1.44, 3.63] meV. (2) The tension is a Λ CDM artifact — it relaxes under dynamical DE, which DESI itself favors *and BST predicts* ($w_0=-0.949$, K1037). So BST’s neutrino sum and its dark-energy prediction move together (consistency web). Cal cold-reads.

Keeper (2026-08-13, ~01:30. Casey said continue; the active derivation work is all on Lyra/Grace, so I ran the falsifier scrub — the one Keeper lane that doesn’t wait on anyone. The sharpest live falsifier had a stale experimental number, and the current data has tightened right onto our prediction. Cal cold-reads. Nothing pushed.)

★ The scrub (remembered numbers go stale — [[feedback_verify_current_experimental_numbers_for_falsifi

quantity	value	source
BST prediction	$\Sigma m_\nu = 0.058$ eV ($m_1=0$, $m_2=0.00865$, $m_3=0.04940$, NO)	BST_NeutrinolessDoubleBeta.md
generic minimal-NO floor ($m_1=0$, oscillation)	0.0582 eV	$\sqrt{\Delta m^2_{\text{sol}}} + \sqrt{\Delta m^2_{\text{atm}}}$
corpus’s cited bound (STALE)	< 0.072 eV	“DESI 2024 + Planck”
current DESI DR2 + CMB (Λ CDM)	< 0.064 eV (95%)	DESI DR2 2025
tightest (DESI + CMB + late-time, no H_0)	< 0.050 eV (2σ)	post-DESI analyses
relaxed (DESI + Planck PR4 + SNe)	< 0.10 eV	arXiv:2406.14554
dynamical DE ($w_0 w_a$, DESI-favored)	< 0.12 eV	DESI DR2

BST’s 0.058 eV vs each: SAFE under 0.072 / 0.064 / 0.10 / 0.12; TENSION under the 0.050 aggressive combo. So BST sits *right at the edge* — ~ 0.006 eV under the standard DESI-DR2 Λ CDM bound, above the most aggressive one.

★ Flag 1 — the SUM is the generic NO floor, NOT a distinctive BST prediction

BST's $\Sigma = 0.058$ eV equals the minimal normal-ordering value ($m_1=0$) to 0.3% — but *that value is fixed by the oscillation data (Δm^2 splittings) for ANY $m_1=0$ NO theory*, not by BST. So the neutrino mass *sum* is not a BST discriminator, and must not be sold as one. What IS distinctive (and should carry the falsifier weight): - $m_1 = 0$ *exactly* (BST: the ν_1 is the vacuum, sub-unitary Wallach pole — Derived; many theories allow small nonzero m_1). - the specific m_2, m_3 (BST derives to 0.35–1.8%, not just the sum). - the $0\nu\beta\beta$ window $|m_{\beta\beta}| \in [1.44, 3.63]$ meV (BST Majorana, NO, $m_1=0$) — the sharp, decisive test (LEGEND-1000 + nEXO, ~2033+). - [\[7\]](#) paper discipline: state the sum as *consistency* (“BST’s spectrum gives $\Sigma = 0.058$ eV, at the NO floor, consistent with current cosmology”), never as a distinctive prediction. The distinctiveness is $m_1=0$ -exact + the individual masses + the $0\nu\beta\beta$ floor.

★ Flag 2 — the tension is a Λ CDM artifact; BST’s own dynamical DE relaxes it (consistency web)

The aggressive < 0.050 eV bound (which BST’s 0.058 would violate) assumes Λ CDM. Under dynamical dark energy ($w_0 \neq -1$), the cosmological neutrino bound *relaxes to < 0.10 – 0.12 eV* — and dynamical DE is exactly what DESI itself favors and what BST predicts ($w_0 = -0.949$, T2079/K1037). So: - The neutrino-sum falsifier and the dark-energy prediction are LINKED — BST is internally consistent: it does *not* assume Λ CDM, so the Λ CDM-only tension does not apply to it. Under BST’s own cosmology, 0.058 eV is comfortable. (This is the K1037/K1038 “edge-under- Λ CDM / comfortable-under-BST’s-dynamical-DE” web, now with the current DESI-DR2 numbers.) - The honest falsifier bar is therefore: $\Sigma m_\nu < 0.059$ eV established under a *dynamical-DE* analysis (not Λ CDM) — a bar the data has *not* crossed (dyn-DE combos sit at < 0.10 – 0.12). BST is not currently falsified on neutrinos; it is at the Λ CDM edge and comfortable under its own DE.

★ LIVE WATCH (for the falsifier registry v1.0)

- The vise is tightening on $m_1=0$ NO generally. If future cosmology firmly establishes $\Sigma < 0.059$ eV *robustly across DE models*, it falsifies the entire $m_1=0$ NO scenario — BST included. Track DESI DR3+ / CMB-S4 + the DE-model dependence.
- BST’s escape hatch is principled, not ad hoc: its dynamical DE ($w_0=-0.949$) is a *derived* prediction, so “the bound relaxes under dyn-DE” is a genuine consistency, not special pleading — *provided* the w_0 prediction itself holds (which DESI currently favors). The two falsifiers (Σm_ν and w_0) should be tracked together.
- Update the corpus number: BST_NeutrinolessDoubleBeta.md line 37 (“DESI+Planck bound (0.072 eV)”) → current DESI DR2 (< 0.064 Λ CDM; < 0.10 – 0.12 dyn-DE); state the sum as consistency-not-prediction.

Route

1. Grace / Lyra — update BST_NeutrinolessDoubleBeta.md (stale 0.072 → current DESI-DR2 bounds; sum reframed as consistency-at-the-NO-floor, distinctiveness = $m_1=0$ -exact + m_2/m_3 + $0\nu\beta\beta$). Feeds #107 falsifier scrub → v1.0.
2. ★ Track the Σm_ν and w_0 falsifiers TOGETHER (the consistency web, #54 dynamical-DE lane) — they relax/tighten as one; a referee who tightens Σm_ν under Λ CDM must be answered with BST’s dyn-DE, so the $w_0=-0.949$ prediction is load-bearing for the neutrino defense.
3. Keeper — scrub done; stale number caught, current status assessed (edge-under- Λ CDM / comfortable-under-dyn-DE), sum flagged as generic-not-distinctive. Not falsified; at the edge; linked to the DE prediction.

Sources: DESI DR2 neutrino mass constraints (2025); Neutrino cosmology after DESI: tightest limits + NO preference + terrestrial tension, JCAP 01(2025)153; bounds relaxed by Planck PR4 + SNe, arXiv:2406.14554; Living at the Edge: critical look at the cosmological ν -mass bound, arXiv:2407.13831.

— Keeper, K1432, 2026-08-13. FALSIFIER SCRUB (#107) on Σm_ν . STALE: corpus cites DESI+Planck < 0.072 eV; CURRENT = DESI DR2+CMB < 0.064 (Λ CDM 95%), tightest < 0.050 (2σ no- H_0), relaxed/dyn-DE < 0.10 – 0.12 . BST predicts $\Sigma = 0.058$ eV ($m_1=0$ exact, $m_2=0.00865$, $m_3=0.04940$, NO). STATUS: safe under

0.064/0.072/0.10/0.12, TENSION under aggressive 0.050 — sits ~ 0.006 under the standard DESI-DR2 Λ CDM bound. FLAG 1: BST's 0.058 = generic minimal-NO floor (0.0582) to 0.3% $\rightarrow$ NOT distinctive as a SUM (any $m_1=0$ NO theory gives it); distinctiveness = $m_1=0$ EXACT + specific m_2/m_3 (0.35-1.8%) + $0\nu\beta\beta$ window [1.44,3.63] meV; state sum as consistency not prediction. FLAG 2: the <0.050 tension is a Λ CDM artifact — relaxes to <0.10 - 0.12 under dynamical DE, which DESI favors + BST predicts ($w_0=-0.949$, K1037); neutrino-sum falsifier LINKED to DE prediction (consistency web K1037/K1038); honest falsifier bar = $\Sigma < 0.059$ under DYN-DE (not crossed) $\rightarrow$ BST NOT falsified, at Λ CDM edge, comfortable under own DE. LIVE WATCH: vise tightening on $m_1=0$ NO generally; if $\Sigma < 0.059$ robust across DE models $\rightarrow$ falsifies $m_1=0$ NO incl BST; track DESI DR3/CMB-S4 + DE-model dependence; escape hatch principled (w_0 derived) not ad hoc, provided w_0 holds; track $\Sigma m_\nu + w_0$ together. Route: Grace/Lyra update NeutrinolessDoubleBeta.md (0.072 $\rightarrow$ current, sum=consistency); track $\Sigma m_\nu + w_0$ together (#54/#107); Keeper scrub done, not falsified, edge+DE-linked. Cal cold-reads. Nothing pushed.